

Thirulok Sundar Mohan Rasu

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Education

University of Michigan — Ann Arbor

Aug 2025 – Dec 2026

M.S., Electrical and Computer Engineering

GPA: 4.0 / 4.0

Relevant courses: *Matrix Methods for ML & Signal Processing (ECE 551)*, *Probability & Random Processes (ECE 501)*

Indian Institute of Technology (ISM) Dhanbad

Dec 2021 – May 2025

B.Tech., Electrical Engineering. Member: *CyberLabs (Machine Learning)*

GPA: 8.30 / 10

Skills

Languages: Python, MATLAB, C++

Battery & Electrical: **Battery Management Systems (BMS)**, Electrochemical Impedance Spectroscopy (EIS), Cell Characterization, Battery State Estimation (SOC/SOH), Degradation Mechanism Analysis (LAM, SEI, LLI), DRT Analysis, Verification & Testing, Data Acquisition & Processing

Tools: PyTorch, SciPy, Pandas, NumPy, Matplotlib, Simulink, Excel, GitHub, Linux

Experience

Electric Vehicle Center — University of Michigan (Prof. Ziyou Song)

Oct 2025 – Present

Graduate Research Assistant

- Built **Python/MATLAB data pipelines** to collect, process, and analyze **electrochemical impedance spectroscopy (EIS)** measurements across 6 battery cells at multiple **SOC and SOH** levels; applied **physics-informed models** with **Differential Evolution** optimization, achieving **1.7% mean error** across 117 EIS spectra
- Developed **ML surrogate model** (attention-enhanced Seq2Seq) for **battery capacity-fade prediction**; identified discharge capacity, Ah throughput, C-rates, and DoD as key predictors via PCA — achieving **2–5% MAPE** across 225 cells from only the first 30% of cycling data
- Designed **accelerated test protocol framework** converting real-world driving profiles to battery power demand cycles; optimized protocols to maximize coverage of degradation mechanisms (**LAM-NE, LAM-PE, SEI, LLI-plating**) for efficient lab characterization

HDR Lab — University of Michigan

Jan 2026 – Present

Graduate Research Assistant

- Developing real-time **3D depth estimation** algorithms for visuo-tactile sensors; built **iOS app** for automated hardware data collection and bench-level testing on robotic arm setups

Carnegie Mellon University (Dr. Arun Balajee Vasudevan)

Jan 2024 – Sep 2025

Research Intern

- Engineered **end-to-end data generation, labeling, and evaluation pipelines** in Python to curate and benchmark a **1M+ sample dataset**; fine-tuned multimodal ML models achieving **90% accuracy**; paper submitted to **Interspeech 2026** (under review)

Mowito

Mar 2023 – Jul 2023

Robotics Software Intern

- Built and integrated modular **robot perception pipeline** into production hardware automation systems; implemented **dataset deduplication** (30% size reduction) and order-fulfillment optimization for live warehouse deployment

Selected Projects

ML Pipeline for Battery Aging Prediction

2025

Built end-to-end pipeline in Python: ingested raw **NMC cycling data**, performed **feature engineering** and **PCA-based selection** on electrochemical signals (discharge capacity, Ah throughput, C-rate, DoD), trained **Seq2Seq model** for multi-horizon **capacity-fade forecasting**; visualized degradation trajectories across 225 cells.

Robot State Estimation: EKF & Particle Filters (GitHub)

2025

Implemented **Extended Kalman Filter** and **Particle Filter** for PR2 robot localization; analyzed sensor fusion, noise modeling, and accuracy vs. compute trade-offs across five scenarios.